CO 2 – ASSESSMENT TOOL -2

COURSE CODE: **DSA0308**

COURSE NAME: **NLP**

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Q1. Morphological processing: analyzing, analysis, analytical

Aim: Identify roots and affixes, classify inflectional and derivational forms, and normalize related words for search engine indexing.

Python program:

```
words = ["analyzing", "analysis", "analytical"]


morphology = {
   "analyzing": {
      "root": "analyze",
      "prefix": "None",
      "suffix": "-ing",
      "type": "Inflectional",
      "normalized": "analyze"
   },
   "analysis": {
      "root": "analy",
      "prefix": "None",
      "suffix": "-sis",
      "type": "Derivational (irregular form)",
      "normalized": "analyze"
   },
   "analytical": {
      "root": "analysis",
      "prefix": "None",
```

```
        "suffix": "-ical",
        "type": "Derivational",
        "normalized": "analyze"
    }
}

print(f"{'Word':<15}{'Root':<15}{'Suffix':<12}"
      f"{'Type':<30}{'Normalized'}")

for word in words:
    item = morphology[word]
    print(f"{word:<15}{item['root']:<15}"
          f"{item['suffix']:<12}{item['type']:<30}"
          f"{item['normalized']}")
```

OUTPUT:

```
>>>
= RESTART: C:/Users/kakar/AppData/Local/Programs/Python/Python314/Co2 at2 q1.py
Word           Root           Suffix      Type                           Normaliz
ed
analyzing      analyze        -ing        Inflectional                   analyze
analysis       analy          -sis        Derivational (irregular form)  analyze
analytical     analysis       -ical       Derivational                   analyze
```

Q2. Morphological parsing: disagree, agreement, agreeable

Aim: Identify prefixes, roots, suffixes, derivational types, and semantic changes for sentiment analysis.

Python program:

```
words = ["disagree", "agreement", "agreeable"]


morphology = {
   "disagree": {
      "prefix": "dis-",
      "root": "agree",
      "suffix": "None",
      "type": "Prefix derivation",
      "meaning": "To have a different or opposing opinion",
      "normalized": "agree"
   },
   "agreement": {
      "prefix": "None",
      "root": "agree",
      "suffix": "-ment",
      "type": "Derivational",
      "meaning": "The state or result of agreeing",
      "normalized": "agree"
   },
   "agreeable": {
      "prefix": "None",
      "root": "agree",
      "suffix": "-able",
      "type": "Derivational",
      "meaning": "Pleasant or acceptable",
      "normalized": "agree"
   }
```

```
}

print(f"{'Word':<15}{'Prefix':<10}{'Root':<10}"
      f"{'Suffix':<10}{'Type':<22}{'Normalized'}")

for word in words:
    item = morphology[word]
    print(f"{word:<15}{item['prefix']:<10}"
          f"{item['root']:<10}{item['suffix']:<10}"
          f"{item['type']:<22}{item['normalized']}")

    print("Meaning:", item["meaning"])
```

OUTPUT:

```
>>>
    = RESTART: C:/Users/kakar/AppData/Local/Programs/Python/Python314/CO2 AT2 Q2.py
    Word           Prefix    Root      Suffix    Type                  Normalized
    disagree       dis-      agree     None      Prefix derivation     agree
    Meaning: To have a different or opposing opinion
    agreement      None      agree     -ment     Derivational          agree
    Meaning: The state or result of agreeing
    agreeable      None      agree     -able     Derivational          agree
    Meaning: Pleasant or acceptable
>>>
```